

Online Appendix

Firm Learning and Log-Normal Distribution: Evidence from Foreign Direct Investment in Viet Nam

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July, 2025 (draft)

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A. Data

A.1. Datasets

A.1.1. Viet Nam Enterprise Survey (VNES)

Data obtained from Mr. Nguyen Viet Phong, General Statistic Office of Viet Nam. The survey is conducted every year by government officials, with rules change from to year. While there is no one official record of the detailed rules for all years, I have managed to extract information from the Vietnamese government’s official websites and related official guidelines. The information to each year’s data collection method from 2010 up to 2025 are in the replication package (project/data/original_paperworks file). The general rules are:

- Time of survey is around March to May each following year of the year surveyed (for example, 2024 data is recorded in April-May of 2025)
- Most registered firms are surveyed and reported, except the following three sectors: sector O (Communist-related entities), sector U (national defense-related entities), and sector T (maid services, household activities to provide for their own use)
- Choice of sample changes from year to year, with different guidance administered by the government annually. For example, 2015 requires all firms with physical presence on Vietnamese land to fill out the survey, while 2020 commits to all big firms and foreign-related firms but only sampled small domestic firms (firms owned by Vietnamese with less than 100 workers).
- There has been a transition of who conduct the survey. Originally, the Ministry of Planning and Investment handled the survey up until the GSO was established and took over the work around 2010. This might explain the hardship in obtaining pre-2011 related documents.

A.1.2. Area code

- Area codes are not consistent over time due to a major code change in 2004, and minor merging/separating of areas throughout the 2000-2018 period. In my dataset, pre-2004 code is changed to be consistent with post-2004 code. Minor changes are not reflected in the dataset: provinces are kept inconsistent if it was merged or separated.
- Provincial code, district code, and ward code (post-2004) are provided by the courtesy of Mr. Nguyen Viet Phong from the Statistic Office of Viet Nam. The file can be accessed using the file *project/data/macro_data/population.xlsx/district.id* in the replication package.
- Provincial code (pre- and post-2004) can be publicly accessed from Viet Nam’s (General Statistic Office, [n.d.](#)), which is only available in Vietnamese, but the interface is fairly simple.

A.1.3. Sector code

- Sector code underwent a major change in 2007.
- Sector code post-2007 are extracted from the government’s legal document ID 10/2007/*ND–CP*, which is provided in *project/data/original_paperworks* in the replication package.
- Sector code pre-2007 are provided by the courtesy of Dr. Brian McCaig, Professor in Economics at Wilfrid Laurier University (as of December 2024). A version of it can be found in *project/data/macro_data/sector.xlsx/sec_div_old* in the replication package.

A.1.4. Inflation and exchange rate

- Inflation data is obtained from GDP deflator from World Bank, [n.d.-a](#).
- Annual exchange rate (VND/USD) is publicly available at World Bank, [n.d.-b](#).
- Vietnamese population is publicly available at World Bank, [n.d.-c](#).

A.1.5. Other datasets

- Province’s educational level, population, urbanization are obtained from National Statistics Office of Vietnam, [n.d](#).
- Viet Nam’s national input-output table is published by the Asian Development Bank (ADB Data Library, [n.d](#)).

A.1.6. Other information

- The FDI-related law are public (in Vietnamese language), obtained from legal documents sourced from various government websites.
- Accounting rules are public (in Vietnamese language), obtained from the Ministry of Finance.
- To get the detailed sources for this paper, please request a collaboration with [ngoclinh_168](#) on Zotero.

A.2. Data choice for empirical observations

The data choice for the empirical observations only include 2009-2015 instead of the full-length dataset. This is because of the couple of reasons:

- I wanted to calculate the productivity using both inputs and investments. For inputs (*cogs* variable), the data was not collected before 2009. For investment (*investment* variable), it is obtained from the original value of fixed capital at the end and beginning

of the year; however, after 2015, that data is either missing or at 0 for more than half of the dataset. Since the calculation of productivity needs accurate and consistent data collecting, such crop in data is important.

- Another reason is the input-output table. The IO data is only available in 2000 and from 2007 on.
- Only manufacturing and service are examined. There is no significant difference in the productivity distribution in agriculture. Furthermore, most agriculture "firms" are family-owned and not surveyed, which means the agriculture data is majorly missing.

B. Proof for propositions

This section will use the zero-profit cutoff (ZPC) and free-entry (FE) conditions intensively. Just a reminder that for all $i = 1, 2, \dots, I$,

$$\frac{1}{\epsilon} r(\varphi_i^*) = w f_i \quad (1)$$

$$M_{ei} \begin{cases} > 0, & \frac{1}{\delta_i} [\frac{1}{\epsilon} r(\tilde{\varphi}_i) - w f_i][1 - G_i(\varphi_i^*)] = w f_{ei} \\ = 0, & \frac{1}{\delta_i} [\frac{1}{\epsilon} r(\tilde{\varphi}_i) - w f_i][1 - G_i(\varphi_i^*)] < w f_{ei} \end{cases} \quad (2)$$

B.1. Proposition 1

Proposition: If $M_1 > 0$, then there exists $i \neq 1$ such that $M_i > 0$.

Proof. Assume only type 1 exists, then $M = M_1 > 0$ and $g_i(\varphi) = g_1(\varphi)$ for all i by construction. Using the FE condition, this means for all $i \neq 1$

$$\begin{aligned} \frac{1}{\epsilon} r(\tilde{\varphi}_1) &= \frac{w \delta_1 f_{e1}}{1 - G_1(\varphi_1^*)} + w f_1 \\ \frac{1}{\epsilon} r(\tilde{\varphi}_i) &< \frac{w \delta_i f_{ei}}{1 - G_1(\varphi_i^*)} + w f_i \end{aligned}$$

which means for all $i \neq 1$,

$$\frac{r(\tilde{\varphi}_1)}{r(\tilde{\varphi}_i)} > \frac{\frac{\delta_1 f_{e1}}{1 - G_1(\varphi_1^*)} + f_1}{\frac{\delta_i f_{ei}}{1 - G_1(\varphi_i^*)} + f_i} \geq \frac{\frac{\delta_1 f_{e1}}{1 - G_1(\varphi_1^*)} + f_1}{\frac{\delta_1 f_{e1}}{1 - G_1(\varphi_i^*)} + f_i}$$

with the last inequality true due to $\delta_i f_{ei} \leq \delta_1 f_{e1}$ in the setup. We will complete this proof by contradiction. In particular, we will show that for all $i \neq 1$,

$$\frac{r(\tilde{\varphi}_1)}{r(\tilde{\varphi}_i)} \leq \frac{\frac{\delta_1 f_{e1}}{1 - G_1(\varphi_1^*)} + f_1}{\frac{\delta_1 f_{e1}}{1 - G_1(\varphi_i^*)} + f_i} \quad (3)$$

Using competitive equilibrium condition $\frac{r(\tilde{\varphi}_1)}{r(\tilde{\varphi}_i)} = \left(\frac{\tilde{\varphi}_1}{\tilde{\varphi}_i} \right)^{\epsilon-1}$, equation (3) becomes

$$\left(\frac{\tilde{\varphi}_1}{\tilde{\varphi}_i} \right)^{\epsilon-1} \leq \frac{\frac{\delta_1 f_{e1}}{1 - G_1(\varphi_1^*)} + f_1}{\frac{\delta_1 f_{e1}}{1 - G_1(\varphi_i^*)} + f_i}$$

which is equivalent to

$$\frac{\delta_i f_{ei}}{(\tilde{\varphi}_i)^{\epsilon-1} [1 - G_1(\varphi_i^*)]} + \frac{f_i}{(\tilde{\varphi}_i)^{\epsilon-1}} \leq \frac{\delta_1 f_{e1}}{(\tilde{\varphi}_1)^{\epsilon-1} [1 - G_1(\varphi_1^*)]} + \frac{f_1}{(\tilde{\varphi}_1)^{\epsilon-1}}$$

Using ZPC condition $\frac{f_1}{f_i} = \frac{r(\varphi_i^*)}{r(\varphi_1^*)} = \left(\frac{\varphi_1^*}{\varphi_i^*}\right)^{\epsilon-1}$, the above is equivalent to

$$\underbrace{\frac{\delta_1 f_{e1}}{(\tilde{\varphi}_i)^{\epsilon-1}[1 - G_1(\varphi_i^*)]} + \frac{f_1}{(\varphi_1^*)^{\epsilon-1}} \left(\frac{\varphi_i^*}{\tilde{\varphi}_i}\right)^{\epsilon-1}}_{\equiv f(\varphi_i^*)} \leq \frac{\delta_1 f_{e1}}{(\tilde{\varphi}_1)^{\epsilon-1}[1 - G_1(\varphi_1^*)]} + \frac{f_1}{(\tilde{\varphi}_1)^{\epsilon-1}}$$

Let the LHS be $f(\varphi_i^*)$, then the RHS is equal to $f(\varphi_1^*)$. Therefore, the above inequality is true if $f(\varphi_i^*)$ is increasing in φ_i^* , since $\left(\frac{\varphi_1^*}{\varphi_i^*}\right)^{\epsilon-1} = \frac{f_1}{f_i} \geq 1$ by ZPC condition and model setup. We have

$$\begin{aligned} \frac{df(\varphi_i^*)}{d\varphi_i^*} &= -\frac{\delta_1 f_{e1}}{\{(\tilde{\varphi}_i)^{\epsilon-1}[1 - G_1(\varphi_i^*)]\}^2} \left[\frac{d}{d\varphi_i^*} (\tilde{\varphi}_i)^{\epsilon-1}[1 - G_1(\varphi_i^*)] \right] + \\ &\quad + \frac{f_1}{(\varphi_1^*)^{\epsilon-1}} (\epsilon - 1) \left(\frac{\varphi_i^*}{\tilde{\varphi}_i}\right)^{\epsilon-2} \frac{\tilde{\varphi}_i - \varphi_i^* \frac{d\tilde{\varphi}_i}{d\varphi_i^*}}{(\tilde{\varphi}_i)^2} \\ &= \frac{\delta_1 f_{e1}}{\{(\tilde{\varphi}_i)^{\epsilon-1}[1 - G_1(\varphi_i^*)]\}^2} [(\varphi_i^*)^{\epsilon-1} g_1(\varphi_i^*)] + \\ &\quad + \frac{f_1}{(\varphi_1^*)^{\epsilon-1}} (\epsilon - 1) \left(\frac{\varphi_i^*}{\tilde{\varphi}_i}\right)^{\epsilon-2} \frac{\tilde{\varphi}_i + \varphi_i^* \frac{d\tilde{\varphi}_i}{d\varphi_i^*} [1 - G_1(\varphi_i^*)]^{-2} g_1(\varphi_i^*) (\varphi_i^*)^{\epsilon-1} g_1(\varphi_i^*)}{(\tilde{\varphi}_i)^2} \\ &> 0 \end{aligned}$$

This concludes the proof. $\square$

B.2. Proposition 2

Proposition: Let all types have the same σ . When $M_1 > 0$, type 1's share $\frac{M_1}{M}$ and each type's distribution are unique.

Proof. To prove this proposition, we will first find the solution for type 1's related statistics and prove that they are unique, which then pins down unique distributions and outcomes for other types. The following subsections show how to solve for each variable and why they are unique.

B.2.1. Type 1's statistics

Productivity cutoff φ_1^*

Combining ZPC (1) and FE (2) when $i = 1$, we can solve for the cutoff φ_1^* as

$$\frac{\delta_1 f_{e1}}{f_1[1 - G_1(\varphi_1^*)]} + 1 = \left(\frac{\tilde{\varphi}_1}{\varphi_1^*}\right)^{\epsilon-1} \quad (4)$$

This pins down φ_1^* uniquely, as G_1 is known and not dependent on M_i distribution. The proof for uniqueness of φ_1^* is similar to appendix B.1. in Melitz, 2003.

Average productivity $\tilde{\varphi}_1$

Subsequently, using φ_1^* , we get a unique $\tilde{\varphi}_1$ by definition because the functional form type 1's source distribution $g_1(\cdot)$ is fixed:

$$\tilde{\varphi}_1 = \left[\int_{\varphi_1^*}^{\infty} \varphi^{\epsilon-1} \frac{g_1(\varphi)}{1 - G_1(\varphi_1^*)} d\varphi \right]^{\frac{1}{\epsilon-1}}$$

Revenue distribution $r(\varphi_1^*)$, $r(\tilde{\varphi}_1)$

Using the ZPC and FE conditions, we can also identify $r(\varphi_1^*)$ and $r(\tilde{\varphi}_1)$ uniquely as

$$\begin{aligned} r(\varphi_1^*) &= \epsilon w f_1 \\ r(\tilde{\varphi}_1) &= \frac{\epsilon w \delta_1 f_{e1}}{1 - G_1(\varphi_1^*)} + \epsilon w f_1 \end{aligned} \quad (5)$$

B.2.2. Other types' cutoffs $\{\varphi_i^*\}_{i \neq 1}$

From Proposition 1, we know that at least one other type besides type 1 will exist. For type $i \neq 1$ that exists, then it must be true from the ZPC (1) condition that

$$\frac{f_1}{f_i} = \left(\frac{\varphi_1^*}{\varphi_i^*} \right)^{\epsilon-1} \quad (6)$$

A unique φ_1^* will pin down φ_i^* uniquely. Here, we can see that neither technological differences ($\sigma_1 - \sigma_2$ and $\mu_1 - \mu_2$) nor the high type's penetration ($\frac{M_1}{M}$) affects the cutoff of each type. This means that realistically, assuming type 1 is foreign firms and type 2 is domestic firms, a shock in productivity of domestic firms will not induce the entrance of very low productivity domestic firms, and vice versa.

B.2.3. Share of type 1 mass $\frac{M_1}{M}$

Using the FE (2) condition for type i ,

$$r(\tilde{\varphi}_i) = \frac{\epsilon w \delta_i f_{ei}}{1 - G_i(\varphi_i^*)} + \epsilon w f_i$$

Using the competitive equilibrium condition $\frac{r(\tilde{\varphi}_1)}{r(\tilde{\varphi}_i)} = \left(\frac{\tilde{\varphi}_1}{\tilde{\varphi}_i} \right)^{\epsilon-1}$, the above equality becomes

$$\tilde{\varphi}_i^{\epsilon-1} = \left(\frac{\epsilon w \delta_i f_{ei}}{1 - G_i(\varphi_i^*)} + \epsilon w f_i \right) r(\tilde{\varphi}_1)^{-1} \tilde{\varphi}_1^{\epsilon-1} \quad (7)$$

By Appendix D.2., the LHS of (7) is strictly increasing in $\mu_i = \underline{\mu}_i + h_\mu(\frac{M_1}{M}, \underline{\mu}_i, \mu_1)$, which means it is increasing in $\frac{M_1}{M}$. Meanwhile, the RHS of (7) is strictly decreasing in $\frac{M_1}{M}$, as $1 - G_i(\varphi_i^*) = 1 - \Phi\left(\frac{\ln(\varphi_i^*) - \underline{\mu}_i - h_\mu(\frac{M_1}{M}, \underline{\mu}_i, \mu_1)}{\sigma_i}\right)$ is strictly increasing in $\frac{M_1}{M}$. Note that all other variables embedded in this equation are given or already solved for, except $\frac{M_1}{M}$. Therefore, the solution for $\frac{M_1}{M}$ in (7) is unique.

B.2.4. Other variables

Once the share of type 1 firms $\frac{M_1}{M}$ is found, we can pin down the unique distributions $\{g_i\}_{i=1,2,\dots,I}$ that each type draws from. Since φ_i^* is unique, we can then solve for

- unique $\tilde{\varphi}_i$ (using its definition)
- unique $r(\tilde{\varphi}_i)$ (using FE condition)
- unique $\{\pi_i(\tilde{\varphi}_i), \pi_i(\varphi_i^*)\}$ (using $\pi_i(\varphi) = \frac{1}{\epsilon}r(\varphi) - wf_i$ in the stationary competitive equilibrium condition)

for all i . □

B.2.5. Algorithm to find types i that exist

So far, we have proved the existence and uniqueness of the solution whenever $M_1 > 0$, but have not yet explicitly showed how to find the set of types i that exist.

The algorithm is relatively simple. Starting with type $i = 2$, we can use the FE and ZPC similar to equation (7) to find a (unique) hypothetical $\frac{M_1}{M}$. Applying that $\frac{M_1}{M}$ to all others distribution, we can find their $r(\tilde{\varphi}_i)$ and check their FE condition. If for all other types, the FE condition is satisfied, then we have found the solution. If there exists a type where the FE condition fails (where $\frac{1}{\delta_i}[\frac{1}{\epsilon}r(\tilde{\varphi}_i) - wf_i][1 - G_i(\varphi_i^*)] > wf_{ei}$), then we will repeat the process with $i = 3$, and so on.

B.3. Proposition 3

Proposition: Let all types have the same σ and $I = 2$. When $M_1 > 0$, the solution for the stationary competitive equilibrium is unique.

Proof. This proposition is an extension to the previous proposition, and this proof is a continuation of the previous proposition's proof.

B.3.1. Mass of each type M_i

So far, we have solved for $\frac{M_1}{M}$, but not the actual shares. One of the stationary competitive equilibrium conditions is $R = wL$. Since labor L is fixed and wage w is normalized to 1, this means R is solved uniquely. We also have $R_i = M_i r(\tilde{\varphi}_i)$, and so

$$\begin{aligned} R &= M_1 r(\tilde{\varphi}_1) + M_2 r(\tilde{\varphi}_2) \\ &= M_1 \underbrace{\left[r(\tilde{\varphi}_1) + \frac{1 - \frac{M_1}{M}}{\frac{M_1}{M}} r(\tilde{\varphi}_2) \right]}_{\text{known \& unique}} \end{aligned}$$

which pins down the numeric value of unique M_1 , then unique M and M_2 .

B.3.2. Revenue, price, and quantity share R_i, P_i, Q_i

To calculate these, we will simply apply the conditions for the stationary competitive equilibrium. With M_i and $r(\tilde{\varphi}_i)$ known and unique, we can calculate each type's total revenue uniquely

$$R_i = M_i r(\tilde{\varphi}_i)$$

Knowing the unique $\tilde{\varphi}_i$ for each type, we will also know its unique average price and price index using

$$p(\tilde{\varphi}_i) = \frac{\epsilon}{\epsilon - 1} \frac{w}{\tilde{\varphi}_i}$$

$$P_i = M_i^{\frac{1}{1-\epsilon}} p(\tilde{\varphi}_i)$$

and quantity using $q(\tilde{\varphi}_i) = \frac{r(\tilde{\varphi}_i)}{p(\tilde{\varphi}_i)}$ and $Q_i = \frac{R_i}{P_i}$ for all i uniquely. $\square$

B.4. Proposition 4

Proposition: Average revenue $r(\tilde{\varphi}_1)$ of type 1 does not depend on $\{\mu_i\}_{i=1,2,\dots}$ distribution, but only on exit rate δ_1 , fixed costs f_{e1}, f_1 , and σ_1 .

Proof. Using (4), by re-arranging variables and expanding $\tilde{\varphi}_1$, then expanding g_1 , we get:

$$\begin{aligned} \frac{\delta_1 f_{e1}}{f_1} &= \left[\frac{\int_{\varphi_1^*}^{\infty} \varphi^{\epsilon-1} \frac{g_1(\varphi)}{1-G_1(\varphi_1^*)} d\varphi}{(\varphi_1^*)^{\epsilon-1}} - 1 \right] [1 - G_1(\varphi_1^*)] \\ &= \int_{\varphi_1^*}^{\infty} \left(\frac{\varphi}{\varphi_1^*} \right)^{\epsilon-1} g_1(\varphi) d\varphi - \int_{\varphi_1^*}^{\infty} g_1(\varphi) d\varphi \\ &= \int_{\varphi_1^*}^{\infty} \left[\left(\frac{\varphi}{\varphi_1^*} \right)^{\epsilon-1} - 1 \right] \frac{1}{\varphi \sigma_1 \sqrt{2\pi}} \exp \left[-\frac{(\ln \varphi - \mu_1)^2}{2\sigma_1^2} \right] d\varphi \end{aligned}$$

Defining $t = \varphi/\varphi_1^*$, the above expression is equivalent to

$$\frac{\delta_1 f_{e1}}{f_1} = \int_1^{\infty} (t^{\epsilon-1} - f_1) \frac{1}{t \sigma_1 \sqrt{2\pi}} \exp \left[-\frac{(\ln(t) + \ln(\varphi_1^*) - \mu_1)^2}{2\sigma_1^2} \right] dt \quad (8)$$

From equation (8), we have $\ln(\varphi_1^*) - \mu_1$ is dependent on only four parameters: $\delta_1, f_{e1}, f_1, \sigma_1$. By expanding $1 - G_1(\varphi_1^*)$ using its definition and a change of variable $t = \frac{\varphi}{\varphi_1^*}$, we have

$$\begin{aligned} 1 - G_1(\varphi_1^*) &= \int_{\varphi_1^*}^{\infty} \frac{1}{\varphi \sigma_1 \sqrt{2\pi}} \exp \left[-\frac{(\ln(\varphi) - \mu_1)^2}{2\sigma_1^2} \right] d\varphi \\ &= \int_1^{\infty} \frac{1}{t \sigma_1 \sqrt{2\pi}} \exp \left[-\frac{(t + \ln(\varphi_1^*) - \mu_1)^2}{2\sigma_1^2} \right] dt \end{aligned}$$

This means $1 - G_1(\varphi_1^*)$ is also dependent on only $(\delta_1, f_{e1}, f_1, \sigma_1)$, and so is $r(\tilde{\varphi}_1)$ because of (5). Therefore, $r(\tilde{\varphi}_1)$ is independent of μ_1 . $\square$

C. Reverse-calculations for calibration

In the dataset, we can observe and directly calculate the following variables: M_i , R_i , w , L_i , μ_{r_i} , σ_{r_i} .

C.1. Elasticity of substitution ϵ

Using 2000-2005 data, I identified the upperbound of ϵ using the following relationship:

$$\begin{aligned} r(\varphi) = p(\varphi)q(\varphi) &= \frac{\epsilon}{\epsilon-1}w(l_i(\varphi) - f_i) \leq \frac{\epsilon}{\epsilon-1}wl_i(\varphi) \\ \Rightarrow \frac{\epsilon}{\epsilon-1} &\geq \frac{r(\varphi)}{wl_i(\varphi)} \end{aligned} \quad (9)$$

Since r (in our case, it is valueadded to labor) is known and l is known for each firm, I took the median of $\frac{r(\varphi)}{wl(\varphi)}$ (valueadded over total labor pay) over all firms and used it as the upperbound for $\frac{\epsilon}{\epsilon-1}$. The median was used because it is more stable than mean in the data, and in the normal distribution, mean and median are the same. In our dataset, the mean of lowerbound over 2001-2005 period for $\frac{\epsilon}{\epsilon-1}$ is 1.2952744, which means the upperbound for ϵ is approximately 4.39. We used $\epsilon = 4$ for our calibration.

C.2. Wage w

At equilibrium, $R = wL$, which means we can get $w = R/L$ directly from observed data.

C.3. Revenue at special points $r(\varphi_i^*)$, $r(\tilde{\varphi}_i)$

Using the zero-profit condition, we have

$$r(\varphi_i^*) = \epsilon w f_i$$

with f_i already identified above. Furthermore, the stationary competitive equilibrium conditions means

$$r(\tilde{\varphi}_i) = R_i/M_i$$

with R_i, M_i observed.

C.4. Fixed production cost f_i

By integrating the expression in (9) over all firms of each type i , we get:

$$R_i = \frac{\epsilon}{\epsilon-1}w(L_i - M_i f_i) \Rightarrow f_i = (wL_i - \frac{\epsilon-1}{\epsilon}R_i)(wM_i)^{-1}$$

with R_i, L_i, M_i observed, w calculated above, and ϵ imposed above.

C.5. Exit rate δ_{it} for each year

For all years, exit rate δ_{it} is determined by number of firms that exist in year t but not $t + 1$, divided by number of firms in t , the current year¹. I calibrated this using firms' unique taxcode². Intuitively, potential firm looks at the most recent year's exit rate to predict their future exit rate.

C.6. Productivity distribution (μ_i, σ_i)

We will use the distribution of $\log(r(\varphi))$ to estimate the productivity of $\log(\varphi)$. Since φ follows a log-normal distribution, Result D.2 in Appendix D shows that for all $\varphi > \varphi_i^*$,

$$\sigma_i = \frac{\sigma_{r_i}}{\epsilon - 1} \text{ for } i = 1, 2$$

$$\mu_i = \frac{\mu_{r_i} - \ln(r(\varphi_i^*)) + (\epsilon - 1) \ln(\varphi_i^*)}{\epsilon - 1}$$

with μ_{r_i}, σ_{r_i} as the index for the log-normal distribution of r_i . Proof is in Appendix C.2.

C.7. Fixed entrance cost f_{ei}

Using results from the previous part and the zero-profit condition:

$$\begin{aligned} \mu_i &= [\mu_{r_i} - \ln(r(\varphi_i^*))]/(\epsilon - 1) + \ln(\varphi_i^*) \\ &= [\mu_{r_i} - \ln(\epsilon w f_i)]/(\epsilon - 1) + \ln(\varphi_i^*) \end{aligned}$$

which means

$$1 - G_i(\varphi_i^*) = 1 - \Phi\left(\frac{\ln(\varphi_i^*) - \mu_i}{\sigma_i}\right) = 1 - \Phi\left(\frac{\ln(\epsilon w f_i) - \mu_{r_i}}{\sigma_{r_i}}\right)$$

with f_i and μ_{r_i}, σ_{r_i} estimated. Finally, using the free-entry condition, we get f_{ei} satisfying

$$f_{ei} = \frac{1}{w\delta_i} \left[\frac{1}{\epsilon} r(\tilde{\varphi}_i) - w f_i \right] [1 - G_i(\varphi_i^*)]$$

with all values on the RHS estimated.

¹An alternative calculation used a lag of exit rate, but the no-lag method offers a more stable path of fixed costs change over time.

²Although taxcodes are not available for all firms, missing and duplicated taxcodes account for less than 1 percent of total number of firms each year during 2001-2005. Meanwhile, exit rates range from 15 to 40 percent.

D. Other propositions

D.1. Strict monotonicity of $\tilde{\varphi}_i$ with respect to φ_i^*

Proposition: $\tilde{\varphi} = \left[\int_{\varphi^*}^{\infty} \varphi^{\epsilon-1} \frac{g(\varphi)}{1-G(\varphi^*)} d\varphi \right]^{\frac{1}{\epsilon-1}}$ is strictly increasing in φ^* when $\epsilon \geq 1$ and $\varphi \geq 0$.

Proof. Take derivative of $\tilde{\varphi}^{\epsilon-1}$ with respect to φ^* to get

$$\begin{aligned}
\frac{d\tilde{\varphi}^{\epsilon-1}}{d\varphi^*} &= \frac{g(\varphi^*)}{[1-G(\varphi^*)]^2} \int_{\varphi^*}^{\infty} \varphi^{\epsilon-1} g(\varphi) d\varphi - \frac{1}{1-G(\varphi^*)} [(\varphi^*)^{\epsilon-1} g(\varphi^*)] \\
&= \frac{(\varphi^*)^{\epsilon-1} g(\varphi^*)}{1-G(\varphi^*)} \left[\frac{\int_{\varphi^*}^{\infty} \left(\frac{\varphi}{\varphi^*}\right)^{\epsilon-1} g(\varphi) d\varphi}{1-G(\varphi^*)} - 1 \right] \\
&= \underbrace{\frac{(\varphi^*)^{\epsilon-1} g(\varphi^*)}{1-G(\varphi^*)}}_{>0} \underbrace{\left[\frac{\int_{\varphi^*}^{\infty} \left(\frac{\varphi}{\varphi^*}\right)^{\epsilon-1} g(\varphi) d\varphi}{\int_{\varphi^*}^{\infty} g(\varphi) d\varphi} - 1 \right]}_{>0, \text{ because } \left(\frac{\varphi}{\varphi^*}\right)^{\epsilon-1} > 1 \text{ for all } \varphi \in (\varphi^*, \infty) \text{ and } \epsilon \geq 1} \\
&\geq 0
\end{aligned}$$

This concludes the proof. $\square$

We can also prove that when $\epsilon < 1$, the reverse ($\tilde{\varphi}$ is decreasing in φ^*) is true with the same proof.

D.2. Strict monotonicity of $\tilde{\varphi}_i$ with respect to μ_i

Proposition: $\tilde{\varphi} = \left[\int_{\varphi^*}^{\infty} \varphi^{\epsilon-1} \frac{g(\varphi)}{1-G(\varphi^*)} d\varphi \right]^{\frac{1}{\epsilon-1}}$ is strictly increasing in μ when $\epsilon \geq 1$ and $\varphi \geq 0$.

Proof. First, expand $\tilde{\varphi}^{\epsilon-1}$ using the definitions of $G(\cdot)$ and $g(\cdot)$

$$\tilde{\varphi}^{\epsilon-1} = \frac{\int_{\varphi^*}^{\infty} \varphi^{\epsilon-1} \frac{1}{\varphi \sigma \sqrt{2\pi}} \exp \left[\frac{-(\ln \varphi - \mu)^2}{2\sigma^2} \right] d\varphi}{\int_{\varphi^*}^{\infty} \frac{1}{\varphi \sigma \sqrt{2\pi}} \exp \left[\frac{-(\ln \varphi - \mu)^2}{2\sigma^2} \right] d\varphi}$$

Differentiate $\tilde{\varphi}^{\epsilon-1}$ using Leibniz's integral rule³, we get

$$\begin{aligned}
\frac{d\tilde{\varphi}^{\epsilon-1}}{d\mu} &= \frac{[\int_{\varphi^*}^{\infty} g(\varphi)d\varphi][\int_{\varphi^*}^{\infty} \varphi^{\epsilon-1}(\ln \varphi - \mu)g(\varphi)d\varphi] - [\int_{\varphi^*}^{\infty} (\ln \varphi - \mu)g(\varphi)d\varphi][\int_{\varphi^*}^{\infty} \varphi^{\epsilon-1}g(\varphi)d\varphi]}{\sigma^4 [1 - G(\varphi^*)]^2} \\
&= \frac{1}{\sigma^4} \left(\underbrace{\int_{\varphi^*}^{\infty} \varphi^{\epsilon-1}(\ln \varphi - \mu)\gamma(\varphi)d\varphi}_{=\mathbb{E}[\varphi^{\epsilon-1}(\ln \varphi - \mu)|\varphi \sim \gamma]} - \underbrace{\left[\int_{\varphi^*}^{\infty} (\ln \varphi - \mu)\gamma(\varphi)d\varphi \right]}_{=\mathbb{E}[\ln \varphi - \mu|\varphi \sim \gamma]} \underbrace{\left[\int_{\varphi^*}^{\infty} \varphi^{\epsilon-1}\gamma(\varphi)d\varphi \right]}_{=\mathbb{E}[\varphi^{\epsilon-1}|\varphi \sim \gamma]} \right) \\
&= \frac{1}{\sigma^4} \text{Cov}(\varphi^{\epsilon-1}, \ln(\varphi) - \mu | \varphi \sim \gamma) \\
&> 0
\end{aligned}$$

where the notation “ $|\varphi \sim \gamma$ ” means “given that φ has the probability density function

$$\gamma(\varphi) = \begin{cases} \frac{g(\varphi)}{1-G(\varphi^*)}, & \text{when } \varphi \geq \varphi^* \\ 0, & \text{when } \varphi < \varphi^* \end{cases}$$

”. Therefore, $\tilde{\varphi}^{\epsilon-1}$ is strictly increasing in μ , which means $\tilde{\varphi}$ is strictly increasing in μ given $\epsilon > 1$. $\square$

D.3. Log-normality of productivity and revenue

Proposition: If productivity follows a log-normal distribution, then revenue also follows a log-normal distribution.

Proof. Assume $f : (-\infty, \infty) \rightarrow (-\infty, \infty)$ such that for all $x_1, x_2 \in (-\infty, \infty)$,

$$f(e^{x_1}) - f(e^{x_2}) = (\epsilon - 1)(x_1 - x_2) \Rightarrow x_1 = [f(e^{x_1}) - f(1)]/(\epsilon - 1)$$

When X follows a normal distribution $N(\mu, \sigma)$. We have

$$\Phi\left(\frac{x - \mu}{\sigma}\right) = \text{Pr}(X \leq x) = \text{Pr}\left(\frac{f(e^X) - f(1)}{\epsilon - 1} \leq x\right) = \text{Pr}(f(e^X) \leq x(\epsilon - 1) + f(1))$$

This is equivalent to

$$\text{Pr}(f(e^X) \leq t) = \Phi\left(\frac{t - f(1) - \mu(\epsilon - 1)}{\sigma(\epsilon - 1)}\right)$$

which mean $f(e^X)$ follows a normal distribution with mean $\mu(\epsilon - 1) + f(1)$ and standard deviation $\sigma(\epsilon - 1)$.

In our model, $\frac{r(\varphi)}{r(\varphi')} = \left(\frac{\varphi}{\varphi'}\right)^{\epsilon-1}$, with φ follows a log-normal distribution with parameters μ and σ (which are the mean and standard deviation of $\ln(X)$). Take log both sides, let

³Leibniz's integral rule: $\frac{d}{d\mu}[\int f(\varphi, \mu)d\varphi] = \int[\frac{d}{d\mu}f(\varphi, \mu)]d\varphi$ for f continuous on $\varphi \in (\varphi^*, \infty)$ and $\mu \in (0, \infty)$.

$f(\varphi) = \ln(r(\varphi))$ and $x = \ln(\varphi)$, we have $\ln(r)$ also follow a normal distribution with mean $\mu(\epsilon - 1) + \ln(r(1))$ and standard deviation $\sigma(\epsilon - 1)$. $\square$

D.4. Relative entrance cutoff

Proposition: The relative cutoff for different types $\frac{\varphi_i^*}{\varphi_{i'}^*}$ does not depend on the technological differences between types, but only the fixed production cost $\frac{f_i}{f_{i'}}$.

Proof. Using the FE conditions for any two types i and i' , then divide each side by each other, we get

$$\frac{r(\varphi_i^*)}{r(\varphi_{i'}^*)} = \frac{f_i}{f_{i'}}$$

Using the equilibrium condition $\frac{r(\varphi_i^*)}{r(\varphi_{i'}^*)} = \left(\frac{\varphi_i^*}{\varphi_{i'}^*}\right)^{\epsilon-1}$, we can conclude

$$\left(\frac{\varphi_i^*}{\varphi_{i'}^*}\right)^{\epsilon-1} = \frac{f_i}{f_{i'}}$$

$\square$

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